

View Reviews

Paper ID

88

Paper Title

Uncertainty-Aware Multimodal Deep Learning Framework for Stress Detection Using Audio-Visual Signals

Track Name

FMLDS2026

Reviewer #3

Questions

1. Is the paper relevant to the conference?

Relevant

2. Novelty/Originality

Low

3. Technical correctness

Low

4. Presentation quality (e.g. organisation, clarity, references, and English)

Low

5. The references are sufficient and up-to-date

Moderate

7. Comments to the author(s)

Thank you for addressing an important problem. The research reflects a valuable clinical research direction. However, the paper requires more revision before publication. Several issues regarding the paper are listed below.

Weakness:

(1) --Major-- Action Unit gating is missing from the paper:

In the abstract, AU gating is listed as a contribution criterion. However, Section III does not give any description, formula, or architectural detail. Please either provide a complete description, mathematical formulation, or remove it completely from all claims. As mentioned in the abstract, the contribution can't be verified.

(2) --Major-- The proposed model underperforms the baseline model:

UA-HybridNet (Acc: 0.813, AUC: 0.878) is outperformed by CNN-LSTM (Acc: 0.885, AUC:

0.971) on every metric. The authors argue that uncertainty justifies this trade-off, which requires proof. Please show: (a) that the CNN-LSTM baseline is poorly calibrated (report its ECE/MCE), and (b) that applying temperature scaling to the CNN-LSTM does not close the calibration gap while retaining its higher accuracy. Without this, it is unclear why UA-HybridNet is preferable.

(3) --Major-- Artificial actors in RAVDESS and lack of stress data:

RAVDESS is a set of emotional data recorded by human actors, not actual stress data. Further, there are 24 actors who performed emotional expressions for the recordings, not acted on behalf of emotional data recording. Make sure your data is actor-disjoint such that no actor appears in both the training and test sets. Else, the model may learn speaker identity rather than stress.

(4) --Major-- Too small a data size:

The test set has only 150 samples (75 per class). At this scale, the confidence interval will be very large, which remains unreported in the paper. Minor variance in the test set might shift accuracy by several percentage points.

(5) --Major-- Missing experimental details:

Please report the total dataset size, train size, validation size, test size, actor overlap policy, MC Dropout rate, number of training epochs, optimizer, learning rate, and batch size. The submitted paper is not reproducible.

(6) --Minor-- Table II caption states, "the deterministic model gives calibrated confidence estimates" — this directly contradicts the paper's argument. Please correct it to "the deterministic model does NOT give calibrated confidence estimates." Else, the claim contradicts the paper's claim.

(7) --Minor-- Figure 2 reads "Confusion Matrix for ." — the model name is missing. Please complete this caption.

(8) --Minor-- The Uncertainty AUROC of 0.6906 is barely above chance (0.5). It suggests uncertainty estimation is a weak indicator of misclassification. Please discuss the result honestly.

(9) --Minor-- Equations 1–5 are standard from previous work. Please consider replacing those with your own mathematical formulation.

(10) --Minor-- Several grammatical errors reduce readability. The paper would benefit from careful proofreading.

9. Overall evaluation

Strong Reject

Reviewer #5

Questions

1. Is the paper relevant to the conference?

Relevant

2. Novelty/Originality

Low

3. Technical correctness

Moderate

4. Presentation quality (e.g. organisation, clarity, references, and English)

Low

5. The references are sufficient and up-to-date

Moderate

7. Comments to the author(s)

1. There is a major mathematical contradiction between the confusion matrix in Figure 2 and the per-class classification report in Table III.

2. The calculated recall for the stress class from the confusion matrix is approximately 86.7%, but Table III lists it as 76.0%. The calculated recall for the non-stress class from the confusion matrix is 76.0%, but Table III lists it as 87.0%

3. Figure 2 contains the header "Pipeline D Confusion Matrix," but the term "Pipeline D" is never introduced or explained in the manuscript. Also the text caption is also incomplete

4. The abstract and introduction highlight an Action Unit gating system as a core part of the hybrid architecture, but this component is completely missing from the system workflow diagram in Figure 1.

5. The abstract contains noticeable typographical and grammatical errors, such as using the phrase "competitive perform" instead of "competitive performance".

9. Overall evaluation

Accept